Consolidated FAI Governance Boundary Map

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Attribution

This work derives from and formalizes the inter-Self coordination architecture introduced in "Inter-Self Coordination via Shared Substrate / Full Aspect Integration" (Li, April 2026), the third paper in the CKS theory series following "Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems" (Li, April 2026) and "The Instinct/Reasoning Separation Outside the Model" (Li, April 2026).

It does not introduce new axioms. Its sole contribution is to consolidate, in reference form, the twelve boundary cases established across notes D6.02 through D6.13 of the Phase D6 boundary series, so that practitioners and prior-art reviewers can locate any scenario on the complete boundary map without reading each case note individually.

Abstract

Notes D6.02 through D6.13 establish twelve boundary cases that together define the complete scope of the Full Aspect Integration (FAI) prior art. This note, D6.14, consolidates those twelve cases into a single reference map organized by type: Type A (minimum boundaries, D6.02–D6.04), Type B (maximum boundaries, D6.05–D6.07), and Type C (definiteness boundaries, D6.08–D6.13). For each case, the map states the boundary criterion and the key test. The twelve tests are jointly exhaustive: any scenario can be located on the map by applying the tests in sequence. A scenario that passes all minimum tests, remains within all maximum ceilings, and satisfies all six definiteness tests is within the prior art's scope. A scenario that fails any test is at or outside the corresponding boundary. Type C boundaries receive the most detailed treatment because they address the grey zones—FAI versus foil, intra-Self versus inter-Self, temporary versus persistent, cooperative versus competitive, FAI versus merger and acquisition, and bilateral versus population scope—where practitioners and reviewers most need decision criteria. Note D6.15 follows as the Phase D6 final closure note.

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1. Why a consolidated boundary map is needed

A prior-art record for a coordination architecture is only as usable as its boundaries are findable. Notes D6.02 through D6.13 each develop one boundary case in depth, working through the underlying architectural commitment, the closest neighbors, and the decision criterion. That depth is necessary for establishing each boundary. But practitioners assessing a scenario and reviewers evaluating an adversarial claim need to move across all twelve boundaries quickly, without reading twelve separate notes each time.

This note serves that need. It extracts the decision-relevant content from each case note—the boundary criterion and the key test—and presents them in a compact, organized reference format. No new commitments are introduced. Every boundary criterion here traces directly to the source paper (Li, April 2026c) and to the specific case note that defends it.

The map serves two functions simultaneously. For practitioners, it answers the question: which boundary applies to the scenario I am assessing, and what is the test? For prior-art reviewers, it answers the question: which case note establishes the boundary that speaks to this adversarial claim, and what does the boundary criterion say? The compact format—boundary criterion plus key test per case—serves both without requiring separate documents.

2. How to read this map

Each entry below gives four items: the case identifier (D6.XX), the boundary type, the boundary criterion (what architectural commitment defines the edge), and the key test (the single question that determines which side of the boundary a scenario falls on).

The twelve tests are organized in three types. Type A tests establish minimum floors—below the floor, a scenario lacks the structural properties the architecture requires. Type B tests establish maximum ceilings—above the ceiling, a scenario exceeds what the architecture specifies. Type C tests establish definiteness—within the broad space bounded by Types A and B, they distinguish which variant of the architecture a scenario instantiates, and whether it falls within the prior art at all.

A scenario within the prior art’s scope must pass all Type A tests (clear the floor), pass all Type B tests (remain below all ceilings), and pass all six Type C tests (resolve to a compliant variant on each definiteness dimension). Failing any single test places the scenario at or outside the corresponding boundary.

3. Type A — Minimum boundaries

Type A boundaries define the floors below which a scenario lacks the structural requirements to be within the prior art’s scope. Three floors apply: one for the FAI event itself, one for the shared substrate, and one for each participating entity.

D6.02 — Minimum valid FAI event

Boundary criterion:

> The seven-requirement minimum viable governance floor established in D2.37. An FAI event must satisfy all seven requirements to be a compliant event: (1) two or more participating Selves each governed under independent governance authority; (2) a shared substrate constructed per Paper 1’s six commitments; (3) aspects as the unit of exchange; (4) exchange bounded to substrate content with instinct-layer content excluded; (5) human authority available over the shared substrate at all times during the event; (6) dissolution governance records specifying what occurs at the event’s end; and (7) the entire coordination mediated by LLMs operating as substrate mediators under

human-authored orchestration rules.

Key test:

Does the event satisfy all seven floor requirements? YES → compliant FAI event. NO → below the minimum; the scenario is not an FAI event regardless of how closely it resembles one in surface behavior.

D6.03 — Minimum valid shared substrate

Boundary criterion:

Six conditions must be satisfied jointly: the substrate is designed with eventual dissolution as a governance intent (temporary design); it carries all six Paper 1 commitments within its scope; all three governance rights (inspect, modify, override) are exercisable over it by participating governance authorities; aspects serve as the exchange units it carries; exchange is bounded such that instinct-layer content does not enter; and the substrate has a defined hand-off capability specifying what content transfers to participating Selves' home substrates at dissolution.

Key test:

Are all six conditions met? YES → the substrate is a compliant shared substrate within the prior art. NO → the object is not a shared substrate in the FAI sense, regardless of how it is labeled.

D6.04 — Minimum valid FAI participant

Boundary criterion:

Four properties must be present in each participating entity: an aspect-cell structure (cells composing into aspects); distinct DNA and action layers within cells; a Self-level specification authored within a home substrate; and human governance authority over the entity's home substrate.

Key test:

Are all four properties present? YES → the entity is a valid FAI participant. NO → the entity is not a Self in the Paper 2 and Paper 3 architectural sense; its participation in a coordination event does not constitute an FAI event.

4. Type B — Maximum boundaries

Type B boundaries define ceilings above which the architecture's scope is exceeded. Three ceilings apply: one for sharing scope, one for participant cardinality, and one for governance depth.

D6.05 — Maximum sharing scope

Boundary criterion:

The absolute ceiling on what may pass through the shared substrate. The instinct layer—comprising LLM weights and the harness substrate that governs instinct-layer behavior—is always home. It does not exchange across FAI events under any configuration. This ceiling is architectural, not configurable: no governance configuration can place instinct-layer content into the shared substrate's exchange scope.

Key test:

Does any shared-substrate content trace to LLM weights or to a participating Self's harness substrate? YES → exchange bounding violation; the scenario exceeds the maximum sharing scope. NO → within the maximum sharing scope ceiling.

D6.06 — Maximum cardinality

Boundary criterion:

There is no architectural ceiling on the number of Selves that may participate in a single FAI event. The architecture specifies no upper bound on N. The practical governor is governance capacity: as N scales, the governance requirements for configuring and overseeing a single event scale with it. Governance capacity failure at high N is a governance quality finding, not an architectural ceiling exceeded.

Key test:

Does governance capacity accommodate the governance requirements that N-party participation generates? YES → compliant high-N FAI event within the prior art. NO → governance capacity failure, which is a deployment finding about the adequacy of the governance arrangement, not a violation of any architectural ceiling.

D6.07 — Maximum governance depth

Boundary criterion:

There is no architectural ceiling on governance depth within a compliant FAI event. Any depth above the minimum floor established in D6.02 is within scope. Configurations with minimal human intervention beyond floor requirements are architecturally compliant; configurations with intensive, recursive, multi-level human oversight are equally compliant. The architecture places no upper bound on how much governance is exercised.

Key test:

Does the event meet at least the floor established by D6.02? YES → compliant at whatever governance depth the participants choose; there is no upper bound test to apply. NO → the floor test fails; governance depth is not the relevant finding.

5. Type C — Definiteness boundaries

Type C boundaries are the most practically important of the three types. Where Types A and B answer binary questions—is the floor met? is the ceiling exceeded?—Type C answers classification questions within the space bounded by Types A and B. The six definiteness boundaries determine what kind of FAI scenario is present, and whether the scenario is within the prior art's scope at all.

These boundaries arise where surface behavior could be classified multiple ways. Practitioners assessing real coordination arrangements encounter these boundaries most frequently because real arrangements are often designed without reference to the architecture's distinctions, and whether a given arrangement falls within the prior art depends on architectural properties that may not be explicitly labeled.

D6.08 — FAI / foil boundary

Boundary criterion:

The seven-requirement governance floor from D6.02 is also the FAI/foil boundary. Partial governance—satisfying some but not all seven floor requirements—produces the foil, not FAI. The foil is the architecturally significant negative case: it names the class of coordination arrangements that resembles FAI in observable behavior but lacks the governance floor that defines the architecture. Common partial-governance scenarios include: using a shared substrate without independent governance authorities for each participating entity; using aspects as exchange units but without human-authored orchestration rules over the shared substrate; maintaining a dissolution governance record but without the three governance rights exercisable during the event.

Key test:

Are all seven floor requirements met? YES → FAI; the scenario is within the prior art. NO, including the case where some but not all requirements are met → foil; the scenario is not within the prior art's scope as an FAI event. Partial compliance does not produce a reduced-scope FAI; it produces a different thing entirely.

Practitioner note: The most common foil pattern is an arrangement that coordinates across organizational boundaries through a shared medium, with effective human oversight in practice but without the three governance rights preserved as architectural properties. Such arrangements may be well-governed in an operational sense; they are not FAI-compliant because the governance is procedural rather than architectural.

D6.09 — Intra-Self / inter-Self boundary

Boundary criterion:

Governance authority independence, not organizational structure. The question is not whether coordinating parties belong to different organizations, different legal entities, or different departments. The question is whether a single set of governance practitioners holds all three rights over all coordinating aspects. If they do—regardless of how the parties are organizationally structured—the coordination is intra-Self (Paper 2 territory). If they do not—if at least two independent governance authorities each hold the three rights over their respective contributed aspects—the coordination is inter-Self (Paper 3 territory).

Key test:

Does a single set of governance practitioners hold all three rights (inspect, modify, override) over all aspects participating in the coordination? YES → intra-Self coordination; Paper 2 addresses this scenario, not Paper 3. NO → inter-Self coordination; the scenario is in Paper 3 scope provided the remaining tests are met.

Practitioner note: This boundary resolves a common misclassification. Two departments within a single organization that share governance authority over their coordination aspects are conducting intra-Self coordination even though they are organizationally distinct units. Conversely, two teams within a single organization that each maintain independent governance authority over their respective aspects—with neither team able to unilaterally inspect, modify, or override the other's aspects—are conducting *inter-Self coordination even though they share an organizational roof.*

D6.10 — Temporary / persistent boundary

Boundary criterion:

Dissolution governance intent, not duration. The FAI architecture requires that the shared substrate be designed with eventual dissolution as an explicit governance intent, with dissolution governance records specifying what occurs when the event ends. A shared substrate intended for permanent operation—one whose governance records do not address dissolution—is outside the FAI scope regardless of how closely it otherwise resembles the architecture. Duration is not the criterion: a shared substrate in operation for years under a dissolution governance intent is within scope; a shared substrate in operation for hours with no dissolution intent is outside scope.

Key test:

Does governance intend eventual dissolution, and are dissolution governance records present in the shared substrate? YES → temporary FAI scope; the scenario is architecturally within the prior art on this dimension. NO (permanent operation intended, or dissolution records absent) → outside FAI scope on this dimension; the coordination arrangement is a different architectural object.

D6.11 — Cooperation / competition boundary

Boundary criterion:

The authored Dimension 4 configuration in the shared substrate, not the event outcome. Cooperation and competition are variants of the same FAI primitive—they use the same shared substrate, the same aspect exchange mechanism, the same governance floor—differing only in the orchestration rules governing contribution scope and content protection. Which variant is present is determined by what Dimension 4 specifies, not by whether participating Selves end up better or worse off, and not by whether the event's outcomes look cooperative or competitive to an outside observer.

Key test:

What does the Dimension 4 orchestration configuration specify? The variant the configuration names is the variant present, regardless of outcome. An outcome that appears to contradict the configured variant is a governance quality finding—the configuration may have been poorly designed, or outcomes may have diverged from intentions—but it does not reclassify the event as a different variant.

Practitioner note: This boundary is important for prior-art purposes because cooperation and competition may appear to be distinct architectures. They are not. Both are within the prior art's scope as FAI variants. An adversarial claim that competitive inter-Self coordination is a novel architecture distinct from FAI fails at this boundary: both cooperation and competition are governance configurations of the same substrate-mediated primitive.

D6.12 — FAI / merger and acquisition boundary

Boundary criterion:

Governance authority preservation after the shared substrate period. FAI events may involve deep collaboration, substantial content exchange, and lasting influence on each participating Self's subsequent evolution. None of these properties distinguishes FAI from M&A.; The distinguishing criterion is whether, at the event's end, two independent governance authorities remain—each retaining the three rights over their respective home substrates and aspects. If governance authority is preserved for both parties, the arrangement is FAI regardless of depth of collaboration. If one party's governance authority has been absorbed into the other's—whether by agreement, by attrition, or by structural merger—the arrangement is M&A; governance, not FAI.

Key test:

After the shared substrate period, are two independent governance authorities preserved, each holding the three rights over their respective home substrates? YES → FAI; deep collaboration is consistent with the prior art provided governance authority remains independent. NO (one governance authority has absorbed the other) → M&A; governance; the scenario is outside the FAI prior art.

Practitioner note: This boundary addresses a scenario in which a shared substrate is used as a mechanism for one entity gradually to absorb another's governance authority. Even if the shared substrate satisfies all floor requirements during the event, the loss of governance independence at dissolution removes the scenario from FAI scope. The boundary criterion evaluates the post-event state, not the event itself.

D6.13 — Bilateral / population scope boundary (calibrated humility)

Boundary criterion:

There is no threshold separating bilateral FAI (two Selves) from multi-party FAI (N Selves) or population-scope coordination (a network of Selves over time). Claims 1 through 5 governance applies at every scale. The architecture does not recognize a scope threshold above which a different governance regime is required. Population-scale dynamics—collective evolution emerging from accumulated FAI events across many Selves—are the subject of Paper 3's Claim 6 and are within the prior art as described there.

Key test:

Does the coordination use Claims 1–5 governance, with aspects as exchange units, a shared substrate carrying the six Paper 1 commitments, and independent governance authorities for participating Selves? YES → within prior-art scope at any scale, from two Selves to population scope. The scale of the deployment is not a criterion for or against prior-art coverage.

Practitioner note: The calibrated-humility voice applicable here follows from Paper 3's Claim 6 register: population-scale dynamics are within the architectural scope the prior art claims, but specific empirical dynamics at population scale await observation. The prior-art record covers the architectural commitment; it does not claim specific empirical predictions about population-scale behavior.

6. The complete-scope property

The twelve tests together are jointly exhaustive. Any scenario can be located on the boundary map by applying them in sequence: first Type A (does it meet the floor?), then Type B (does it stay within the ceilings?), then Type C (which variant is it, and is it within scope on each definiteness dimension?).

A scenario is within the prior art's scope if and only if: it passes all three Type A tests, passes all three Type B tests, and resolves to a compliant variant on all six Type C dimensions. Any test failure places the scenario at or outside the boundary the failing test governs. Multiple test failures are possible; each identifies a separate boundary the scenario engages.

For prior-art review purposes, the map's structure means that any adversarial claim about a scenario must engage a specific boundary. A claim that a scenario is novel because it is "like FAI but different" must specify which boundary criterion it falls outside of. The twelve tests provide the vocabulary for that specification.

7. Conclusion and series position

This note completes the reference function of the Phase D6 boundary series. Notes D6.02 through D6.13 each establish one boundary case in depth; this note, D6.14, presents all twelve as a consolidated map for practitioner and prior-art use.

Note D6.15 follows as the Phase D6 final closure note, completing the boundary series for Paper 3's inter-Self coordination architecture.

Source papers

Li, W. (2026a). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems.* April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026b). *The Instinct/Reasoning Separation Outside the Model.* April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026c). *Inter-Self Coordination via Shared Substrate / Full Aspect Integration.* April 2026. ORCID: 0009-0004-8065-3235.

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